Marathon Runner: Water Balance — Teacher Guide

A one-screen model of a runner's body during a race. Students choose how she runs and what she drinks, run the race in about a minute, and watch what happens to the water and salt in her blood.

At a glance

Where	sims.mr-miner.com/play/water-balance (backup: sfminer.github.io/water-balance)
Devices	Any browser, including Chromebooks. No login, nothing saved between visits.
Time	A 4-hour race takes about 1 minute at Speed 1x (8 hours: 2 minutes). Speed 4x cuts that to 15–30 s. Plan 25–35 minutes for the worksheet.
Fits	Marathon Runner anchor phenomenon (Water Balance 5E). Best after students have seen the Medical Tent Data, before or during Explain.
Pairs with	The osmosis / cell-size simulation, which shows what happens to the cells themselves.

What students can change

Setting	Choices	What it does
Pace	Stop / Run	Running sweats about 8 times as fast as resting. Legs move on screen while running.
How much the runner drinks	None / Medium / Lots	Medium about matches what she sweats. Lots is more than sweat plus her kidneys can get rid of.
What the runner drinks	Water / Sports drink	A sports drink adds a little salt with the water.
Race length	4 / 6 / 8 hours	Same rates, just longer. 4 hours is the default.

- **Mid-race changes.** Settings can be changed in the middle of a race (for example, start running, then switch to Stop halfway). Changing a setting after a race ends starts a fresh race.

The buttons

Button	What it does
Start race / Pause / Resume	Starts and pauses the race clock. Space does the same.
Run again	Appears when the race is over; runs the same settings from the start.
Restart	Goes back to the start of the race, keeping the settings.
Speed 1x / 2x / 4x	How fast the race clock runs. Results are the same at every speed.
Keep this run	Freezes the current sodium line on the graph as a dashed line, so the next run can be compared with it.
Clear kept run	Removes the dashed line.

Reading the screen

- **The body diagram (left):** cup (drinking), sweat leaving the skin, the blood box, the ADH meter, and the kidneys with the urine arrow. Moving arrows get thicker when more is flowing.
- **The blood box:** the water level shows how much water is in her blood; the yellow diamonds show how much salt. Crowded diamonds = salty blood; spread-out diamonds = diluted blood.
- **ADH meter (purple):** the brain's message to the kidneys. High ADH = keep water (little urine). Low ADH = let water go (more urine).
- **The observation line:** one sentence describing what is happening right now: is the blood saltier or more diluted, is water moving into or out of the cells, and what ADH and the kidneys are doing.
- **Water and salt so far (right):** how much water and salt has gone in (Drunk), out (Sweat, Urine), and how much is in the blood now. Every number is a percentage of what the blood held at the start, so 100% = no change.
- **Blood saltiness, gauge and graph:** salt ÷ water, turned into the mEq/L units on the Medical Tent Data. The green band is the normal range, 135–145 mEq/L.

When the runner collapses

If a choice pushes her body too far, the race stops, her label turns red ("Runner (collapsed)") and the observation line says why: too much water lost, too much water held, sodium too low or sodium too high. This is intended: it keeps the screen from showing values no one could survive. Students can press Run again or change a setting. The collapse points (blood water below 65% or above 150% of its start; sodium below 110 or above 170 mEq/L) are classroom limits, not medical cut-offs.

What to expect (answer key)

Blood sodium at the end of the race, in mEq/L (normal 135–145). Small differences of ±1 can appear.

Race	Pace	None	Medium	Lots of water	Lots of sports drink
4 h	Run	164 (high)	139	125 (low)	132 (low)
4 h	Stop	143	136	113 (low)	122 (low)
6 h	Run	collapses ≈5.0 h	139	121 (low)	132 (low)
8 h	Run	collapses ≈5.0 h	138	117 (low)	132 (low)
8 h	Stop	146	135	collapses ≈4.6 h	collapses ≈6.2 h

The big ideas students should find:

1. **No drinking → too salty.** Running and drinking nothing makes the blood saltier (sweat takes out water). ADH goes up and urine drops: the body saves water, but can't make up for no drinking.
2. **About the right amount → normal.** Medium drinking keeps her in or near the normal range. If she drinks a little extra, ADH falls and the kidneys get rid of it.
3. **Too much water → too diluted.** Drinking lots of plain water dilutes the blood below normal, even with ADH at zero, because she drinks faster than her kidneys can get rid of water. Water then moves into the cells (including brain cells), which is why this is dangerous. This is the anchor phenomenon.
4. **Stopping makes overdrinking worse.** Resting while drinking lots is worse than running while drinking lots, because a resting runner sweats less water away.
5. **Sports drinks help a little.** A sports drink softens the drop but does not prevent it: the problem is mostly too much water, not too little salt.

Watch for these misconceptions

- **"Lower number = she lost more sodium."** A lower sodium number means the blood is more diluted, not necessarily that she lost more salt. Point students to the Salt row: the running runner loses more salt in sweat than the resting runner, yet her blood ends up saltier, because sweat takes out even more water.
- **Amount vs. concentration.** Water in the blood and saltiness of the blood are different things. Use the blood box: same number of diamonds, more water, lower saltiness.
- **"The kidneys should just fix it."** The kidneys do respond, but they have a maximum. With Lots of water, ADH reaches zero and the observation says the kidneys are at their limit.

Good to know

- **It's a model.** The numbers are illustrative: they behave like the real body but are not medical measurements, and the race clock is scaled. The game does not contain the runner's own Medical Tent values; students use the printed data.
- **Keep the tab on screen.** The game pauses itself if its tab is hidden or the window is minimized. Keep the game visible while it runs.
- **Old version showing?** If the game looks like an older version, press Ctrl+Shift+R to reload it.
- **Keyboard.** Tab moves through the buttons (yellow outline); Space starts and pauses; Escape or tabbing past the last button leaves the game for the rest of the page.
- **Small screens.** Text stays readable on small screens; on a phone or a short window, scroll to see the graph.